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Diagnosis-Guided Augmentation and Frequency-Gated Fusion for Actor-Conditioned Human-Value Recognition in News

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A news topic does not identify an actor's values

Chile protests | Reported government statement

> *"He justified the decision, saying he wanted to maintain* ***public order*** *against vandals and crime."*

Whose conduct?	Which value?	Which direction?
Government and Security Forces	Have a safe country	**Aligned**

Benchmark annotation of reported conduct; not a judgment of permanent character.

The actor, value and direction must be interpreted together.

Source excerpt and gold label: Huang, midterm paper, §5.9, Table 9 (Chile case).

Why this matters for news analysis

01 Analyze how institutions are portrayed

Connect reported conduct to the actor involved.

02 Organize value-related news

Find reports about security, fairness and other values.

03 Support evidence-grounded summaries

Preserve who did what, and why a value is relevant.

Actor-specific analysis preserves who did what and which value is involved.

Potential uses, not evaluated deployments: Huang, midterm paper, abstract and §1.1.

The task predicts values for each supplied actor

Input

News article

Supplied actor list

Output for each actor

Value + aligned / contradictory

Multiple labels are allowed.

Task	Label space
Task 1b — main task	54 fine values × 2 directions = 108 labels
Task 1a	20 coarse categories, projected from fine values

At test time, Task 1 uses the article—not official subevents.

Task setting: FEHU overview [1]; Huang, midterm paper, §3.1.

More predicted values can also mean more errors

Challenge	Why it is difficult	Possible error
Implicit evidence	The value is not named in the text.	Missed value
Multiple actors	Their actions and interests differ.	Wrong actor
Few training examples	Infrequent labels are hard to learn.	Poor coverage
Expanding predictions	A plausible value may lack support.	False or reversed label

Recovering a candidate and judging its direction need different evidence.

Challenges and error definitions: Huang, midterm paper, §§1.1, 3.2 and 5.2.

Control recovery, selection and direction separately

Goal: improve actor-specific value recognition with limited, imbalanced supervision.

Design idea	What changes	Expected effect
Targeted examples	Generate for diagnosed label failures	Recover missed values
Frequency-gated fusion	Limit which predictions can be revised	Control false positives
Selective direction QA	Check disagreements after selection	Correct some reversals

So far: development Micro-F1 0.5496 → 0.5817.

Implemented and measured; matched tests of individual contributions remain planned.

Ideas and progress: Huang, midterm paper, §§1.2–1.3, 4 and Table 7.

Three research questions guide the evaluation

RQ1 Where does recognition fail?

Frequency, representation and direction errors

RQ2 What changes with generated training examples?

Targeted data and unspecified-label supervision

RQ3 How does selective fusion change the errors?

Precision, recall, label balance and direction

Diagnosis → repeated training → saved-stage comparisons

Research questions: Huang, midterm paper, §3.3.

Related work uses a different information flow

Approach	Setting or information flow	Difference here
ValueEval [4]	Human values behind arguments	Actors and directions in news
EAVIT [7]	Local detector → reduced prompt → LLM	Encoder supports prediction revision
This study	**Prompt outputs → gated revision → direction QA**	**Separate selection from direction**

Design comparison; no matched performance experiment against EAVIT.

Encoder support revises prompt predictions before the direction check.

Sources: ValueEval [4], EAVIT [7] (ijcai.org/proceedings/2025/934); Huang, §2.

Two experiment tracks answer different questions

Training study

Generated records → encoder training → three-seed evaluation

Value classifier: DeBERTa-v3-base + LoRA [9, 10]; 108 output scores

Inference study

Prompt predictions + historical encoder scores → fusion stages

The archived encoder scores are not linked to a repeated-training checkpoint.

Experimental tracks: Huang, §§1.3, 4.3, 5.4, 5.10 and Appendix A; model choices [9, 10].

Generate training examples for diagnosed failures

METHOD · TARGETED AUGMENTATION

18 target labels: low support and zero baseline recall

Selection uses development diagnosis at prediction threshold 0.30.

Round 2 generation mode	Records (count)
Targeted full examples	450
Direction contrasts	180
Situation transformations	180
Hard-negative requests	90

Round 1: 340 screened records. Round 2: 900 structurally checked records.

Preserve the actor, value and direction in each training example.

Generation design and record counts: Huang, §4.2, Table 1 and Appendix A.

An unspecified label is not a verified negative

A generated record may describe one target without labeling every other value.

Label status	Assume-negative	Masking
Known positive	Train as positive	Train as positive
Known negative	Train as negative	Train as negative
Unspecified	**Train as negative**	**Exclude from loss**

Masking removes an assumption, but also removes negative supervision.

Supervision comparison: Huang, §§4.3 and 5.6; partial-label learning [8].

Revise eligible labels, then check direction

Primary predictions

Article → generated event descriptions → value predictions

Gated revision

Remove weak labels
Add supported labels
Prompt + encoder

Direction check

Revisit selected aligned / contradictory disagreements

38 / 108 labels pass the training-frequency gate (>130 positives).
Other labels keep the primary predictions at these two stages.

All 18 augmentation targets are outside the removal/addition gate.

System flow: Huang, §§4.4–4.7, Figure 3 and Algorithm 1.

Keep data splits and evaluation measures distinct

Split	Articles (count)	Actors (count)
Train	1,228	3,381
Development	177	459
Task 1 test	175	479

Actors = supplied article–actor units.

Encoder repetitions use seeds 42, 43 and 44.

Micro-F1 ↑

Overall directed-label recognition

Macro-F1 ↑

Equal weight per supported label

Direction reversal rate ↓

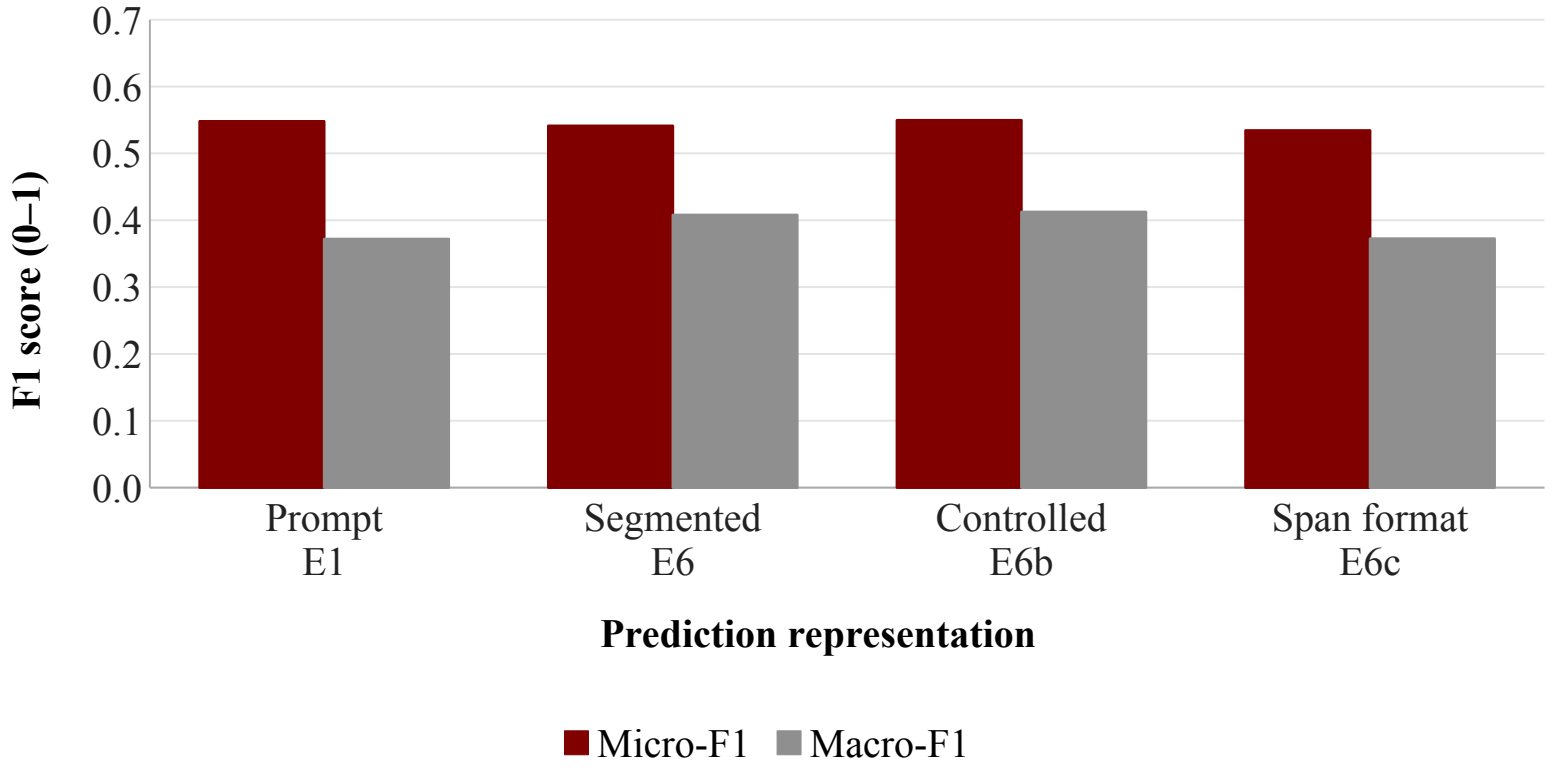
Wrong direction on gold-exclusive pairs

Scores are proportions (0–1). Development choices and official test stay separate.

Dataset counts and metric definitions: Huang, §§5.1–5.3, Table 2.

Segmentation changes the error balance

RQ1 · REPRESENTATION DIAGNOSIS ON DEVELOPMENT



Controlled vs E6

Micro-F1

0.5412 → 0.5496

Reversal rate

0.0504 → 0.0459

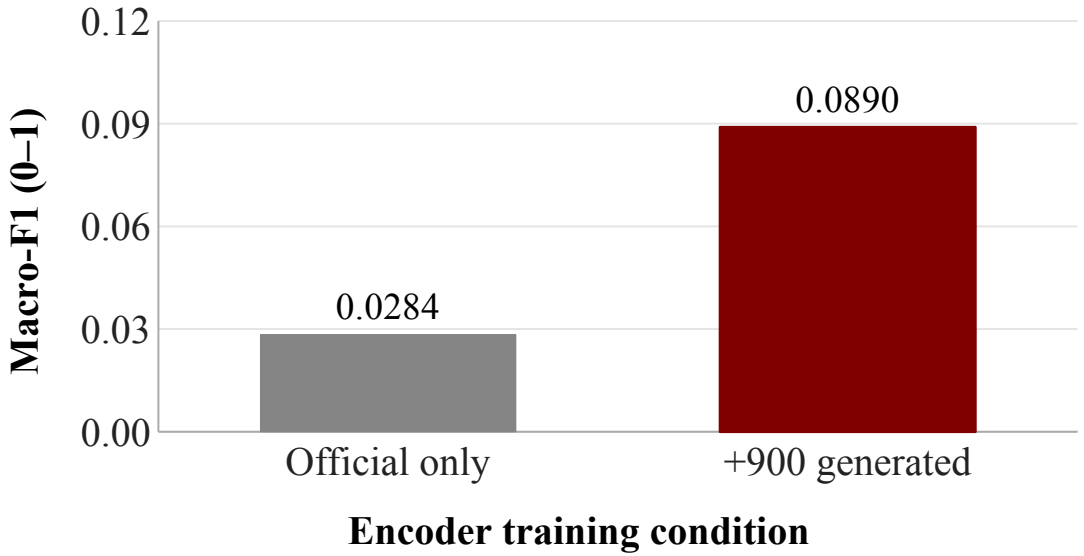
Granularity control helps, while direction errors remain.

Huang, §5.5, Table 4. E1 is the prompt baseline; E6b is the fusion primary system.

Ranking improves; target prediction remains weak

RQ2 · AUGMENTATION RESULTS ON DEVELOPMENT

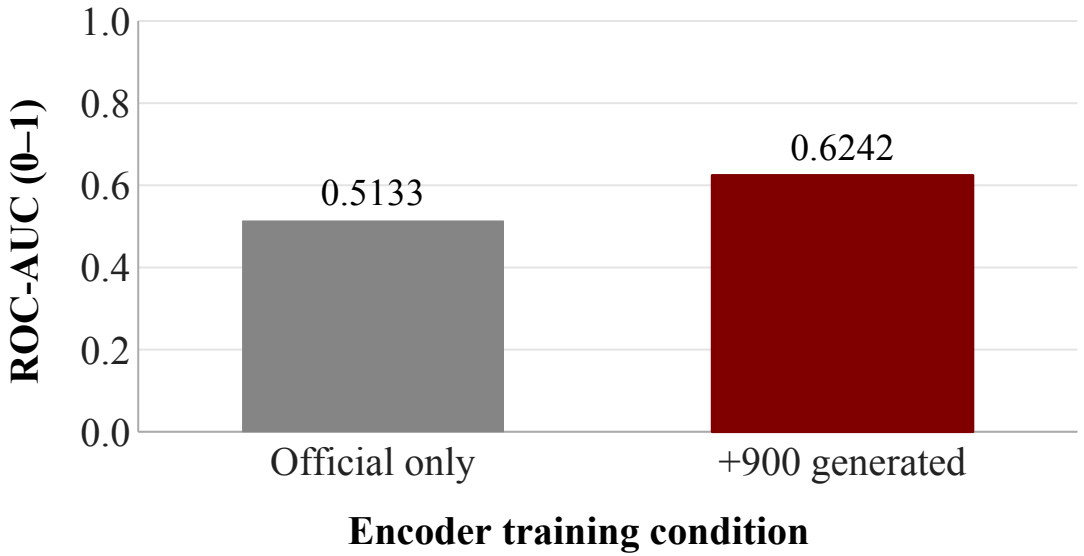
Full-space Macro-F1



Sample SD: 0.0018 / 0.0370

Three-seed means; axis scales differ.

Target ROC-AUC



Sample SD: 0.0304 / 0.0108

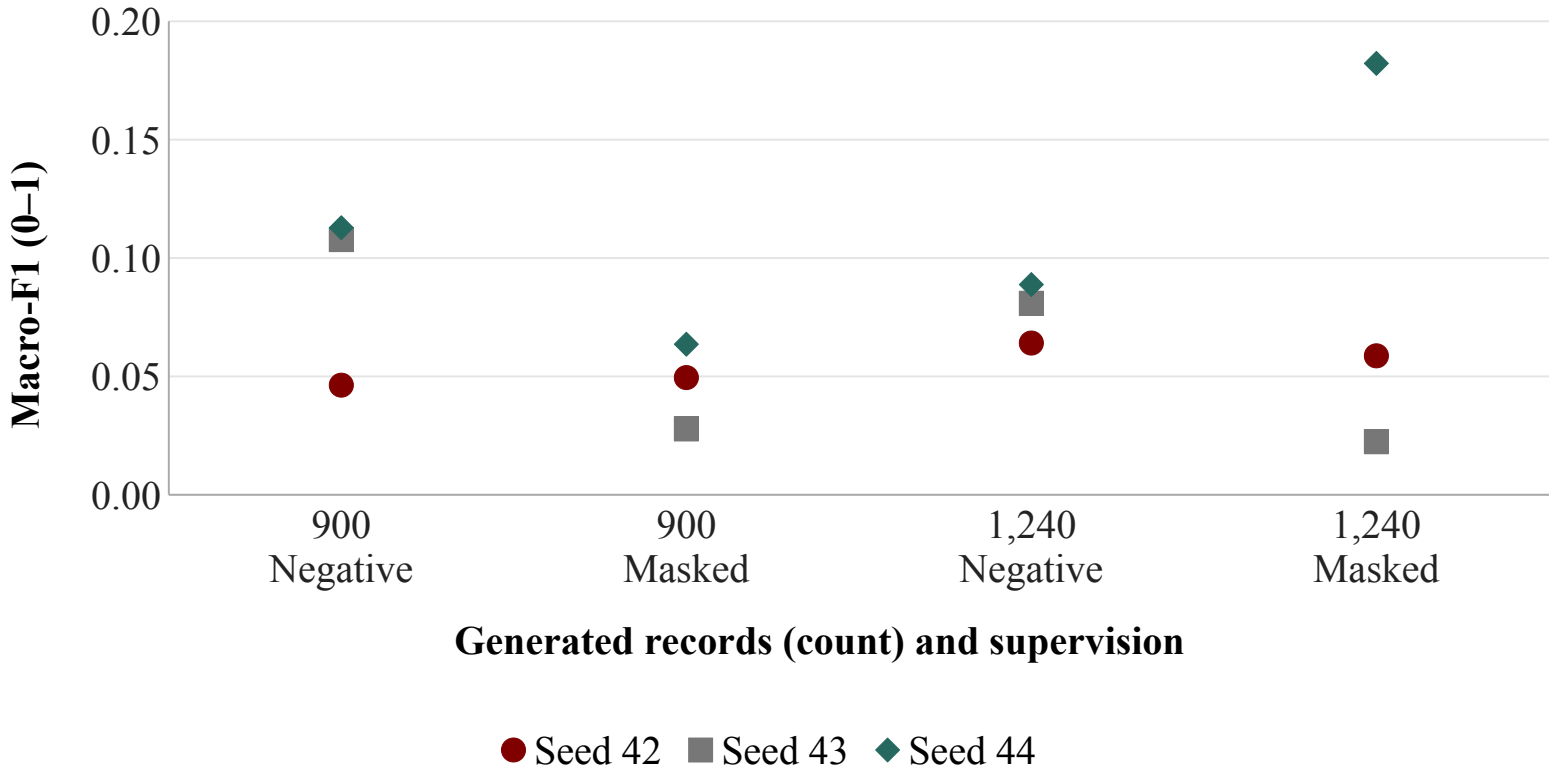
AUC measures ranking, not label accuracy.

Target F1: 0.0000 → 0.0319 at threshold 0.30. Matched data controls are pending.

Huang, §5.6, Table 5 and Figure 5. A vs B: official-only vs 900 assume-negative records.

Masking is sensitive to data and random seed

RQ2 · UNSPECIFIED-LABEL SUPERVISION



Same 900 records

Masking lowers mean Macro-F1.

1,240 masked

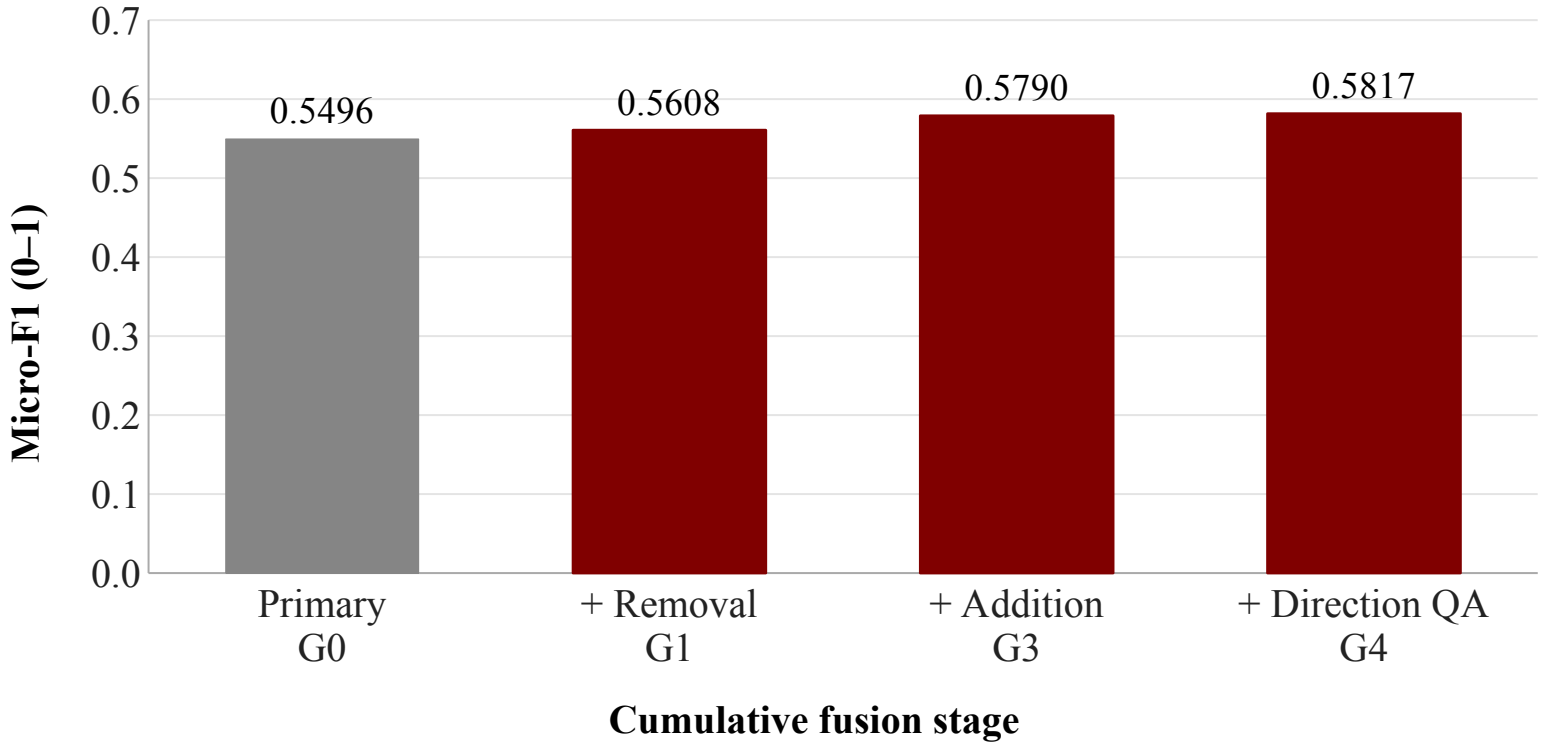
One run drives the higher mean.

Two of three 1,240-record masked runs make no correct target predictions.

Seed values: Huang, §5.6, Figure 5. Negative = treat unspecified labels as negative.

Removal and addition provide most of the gain

RQ3 · FUSION STAGES ON DEVELOPMENT



+3.21 pp

Overall Micro-F1 improvement

95% interval
+2.09 to +4.33
percentage points

Article bootstrap; saved outputs only

Removal raises precision; addition restores recall. Direction QA adds a small gain.

Huang, §5.7, Table 7 and Figure 6. pp = percentage points; G0 is the fixed primary baseline.

The frequency gate protects label balance

RQ3 · REMOVAL-BRANCH COMPARISON

Removal configuration	Micro-F1 ↑	Macro-F1 ↑
Gated removal (G1)	**0.5608**	**0.4146**
Ungated removal (G5)	0.5703	0.3900

Ungated: +0.95 pp Micro-F1 **Ungated: −2.46 pp Macro-F1**

Scores are proportions (0–1); pp = percentage points.

This is a removal-only comparison, not a complete-system gate ablation.

Huang, §5.7, Table 7. G1 and G5 differ within the removal branch.

Official test results show strengths and limits

EVALUATION · TASK 1b OFFICIAL TEST

Submitted system	Micro-F1 ↑	Macro-F1 ↑	DRR ↓
WAIPS-01 — team comparison	0.5543	0.3277	0.0345
WAIPS-02 — simpler comparator	0.5496	0.3580	0.0395
WAIPS-03 — proposed fusion	**0.5808**	**0.3952**	**0.0477**
IIUoT-01 — other participant	0.5979	0.3930	0.0613
Organizer baseline — unranked	0.6284	0.4402	0.0386

All scores: proportions (0–1). DRR = direction reversal rate. Call budgets differ.

Highest participant Macro-F1; official ranking uses Micro-F1.

Official scores reproduced from FEHU overview [1] and WAIPS manuscript [2]; Huang, Table 8.

Direction checks can fix or introduce an error

CORRECTED · Venezuela sanctions

*"The situation will **deteriorate** with these last sanctions especially."*

Actor: US Government and Allied Nations

Value: Have a stable society

Aligned → Contradictory

Gold: contradictory

INTRODUCED ERROR · Chile protests

*"...he wanted to maintain **public order** against vandals and crime."*

Actor: Government and Security Forces

Value: Have a safe country

Aligned → Contradictory

Gold: aligned

A label does not show which intention, action or consequence drove the decision.

Excerpts, actor names and gold labels: Huang, §5.9, Table 9. Ellipsis marks a shortened excerpt.

What the current evidence can establish

Supported by the current experiments	Still needs a stronger test
Generated records improve encoder ranking	Targeting vs equal-sized generic data
Saved fusion stages improve the primary set	Augmented encoder contribution in fusion
Direction QA has a small net benefit	Actor-linked evidence and rerun stability

Development data serves several selection roles; historical checkpoint lineage is incomplete.

The system is implemented; contribution attribution is the main remaining task.

Evidence limits: Huang, §§5.10, 6 and Appendix A.

Next, isolate the contribution of each idea

FUTURE WORK · PRIORITIZED EXPERIMENTS

1 **Fix prompt outputs; change only encoder support**

No encoder / official-only encoder / augmented encoder

2 **Match the size of the added training data**

Targeted generation / generic generation / oversampling

3 **Test reliability before extending the task**

Separate selection split; label audit; evidence and cost evaluation

Planned extensions: actor-linked evidence and Task 2, after the core comparisons.

Future studies: Huang, §§6.1–6.2. All comparisons on this slide are planned.

References: task and value recognition

REFERENCES · COMPLETE CITATIONS IN THE NOTES

[1] Wang et al. Overview of the NTCIR-19 FEHU Task.
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References: methods and evaluation

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Complete bibliography: Huang, submitted midterm paper. Citation numbering matches that document.

Conclusion: recover values and control the errors

Goal: recognize whose conduct supports or contradicts a value when training supervision is limited and imbalanced.

Contribution	Targeted data, frequency-gated revision and separate direction checking
Main evidence	Development Micro-F1: 0.5496 → 0.5817 Official test: 0.5808 Micro-F1 / 0.3952 Macro-F1
Next step	Isolate the gains with matched comparisons and verify actor-linked evidence

Better label recovery needs careful selection and evidence for direction.

Conclusions: Huang, §§5–6; development Table 7 and official test Table 8.